

7D Isometries, Unitary Operators, and Matrix Factorization

Isometries

Linear maps that preserve norms are sufficiently important to deserve a name.

7.44 definition: *isometry*

A linear map $S \in \mathcal{L}(V, W)$ is called an *isometry* if

$$\|Sv\| = \|v\|$$

for every $v \in V$. In other words, a linear map is an isometry if it preserves norms.

If $S \in \mathcal{L}(V, W)$ is an isometry and $v \in V$ is such that $Sv = 0$, then

$$\|v\| = \|Sv\| = \|0\| = 0,$$

which implies that $v = 0$. Thus every isometry is injective.

*The Greek word **isos** means equal; the Greek word **metron** means measure. Thus **isometry** literally means equal measure.*

7.45 example: *orthonormal basis maps to orthonormal list $\Rightarrow$ isometry*

Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $g_1, \dots, g_n$ is an orthonormal list in W . Let $S \in \mathcal{L}(V, W)$ be the linear map such that $Se_k = g_k$ for each $k = 1, \dots, n$. To show that S is an isometry, suppose $v \in V$. Then

$$7.46 \quad v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n$$

and

$$7.47 \quad \|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2,$$

where we have used 6.30(b). Applying S to both sides of 7.46 gives

$$Sv = \langle v, e_1 \rangle Se_1 + \dots + \langle v, e_n \rangle Se_n = \langle v, e_1 \rangle g_1 + \dots + \langle v, e_n \rangle g_n.$$

Thus

$$7.48 \quad \|Sv\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2.$$

Comparing 7.47 and 7.48 shows that $\|v\| = \|Sv\|$. Thus S is an isometry.

The next result gives conditions equivalent to being an isometry. The equivalence of (a) and (c) shows that a linear map is an isometry if and only if it preserves inner products. The equivalence of (a) and (d) shows that a linear map is an isometry if and only if it maps some orthonormal basis to an orthonormal list. Thus the isometries given by Example 7.45 include all isometries. Furthermore, a linear map is an isometry if and only if it maps every orthonormal basis to an orthonormal list [because whether or not (a) holds does not depend on the basis $e_1, \dots, e_n$].

The equivalence of (a) and (e) in the next result shows that a linear map is an isometry if and only if the columns of its matrix (with respect to any orthonormal bases) form an orthonormal list. Here we are identifying the columns of an m -by- n matrix with elements of $\mathbf{F}^m$ and then using the Euclidean inner product on $\mathbf{F}^m$.

7.49 characterizations of isometries

Suppose $S \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Then the following are equivalent.

- (a) S is an isometry.
- (b) $S^*S = I$.
- (c) $\langle Su, Sv \rangle = \langle u, v \rangle$ for all $u, v \in V$.
- (d) $Se_1, \dots, Se_n$ is an orthonormal list in W .
- (e) The columns of $\mathcal{M}(S, (e_1, \dots, e_n), (f_1, \dots, f_m))$ form an orthonormal list in $\mathbf{F}^m$ with respect to the Euclidean inner product.

Proof First suppose (a) holds, so S is an isometry. If $v \in V$ then

$$\langle (I - S^*S)v, v \rangle = \langle v, v \rangle - \langle S^*Sv, v \rangle = \|v\|^2 - \langle Sv, Sv \rangle = \|v\|^2 - \|Sv\|^2 = 0.$$

Hence the self-adjoint operator $I - S^*S$ equals 0 (by 7.16). Thus $S^*S = I$, proving that (a) implies (b).

Now suppose (b) holds, so $S^*S = I$. If $u, v \in V$ then

$$\langle Su, Sv \rangle = \langle S^*Su, v \rangle = \langle Iu, v \rangle = \langle u, v \rangle,$$

proving that (b) implies (c).

Now suppose that (c) holds, so $\langle Su, Sv \rangle = \langle u, v \rangle$ for all $u, v \in V$. Thus if $j, k \in \{1, \dots, n\}$, then

$$\langle Se_j, Se_k \rangle = \langle e_j, e_k \rangle.$$

Hence $Se_1, \dots, Se_n$ is an orthonormal list in W , proving that (c) implies (d).

Now suppose that (d) holds, so $Se_1, \dots, Se_n$ is an orthonormal list in W . Let $A = \mathcal{M}(S, (e_1, \dots, e_n), (f_1, \dots, f_m))$. If $k, r \in \{1, \dots, n\}$, then

$$7.50 \quad \sum_{j=1}^m A_{j,k} \overline{A_{j,r}} = \left\langle \sum_{j=1}^m A_{j,k} f_j, \sum_{j=1}^m A_{j,r} f_j \right\rangle = \langle Se_k, Se_r \rangle = \begin{cases} 1 & \text{if } k = r, \\ 0 & \text{if } k \neq r. \end{cases}$$

The left side of 7.50 is the inner product in $\mathbf{F}^m$ of columns k and r of A . Thus the columns of A form an orthonormal list in $\mathbf{F}^m$, proving that (d) implies (e).

Now suppose (e) holds, so the columns of the matrix A defined in the paragraph above form an orthonormal list in $\mathbf{F}^m$. Then 7.50 shows that $Se_1, \dots, Se_n$ is an orthonormal list in W . Thus Example 7.45, with $Se_1, \dots, Se_n$ playing the role of $g_1, \dots, g_n$, shows that S is an isometry, proving that (e) implies (a). ■

See Exercises 1 and 11 for additional conditions that are equivalent to being an isometry.

Unitary Operators

In this subsection, we confine our attention to linear maps from a vector space to itself. In other words, we will be working with operators.

7.51 definition: unitary operator

An operator $S \in \mathcal{L}(V)$ is called *unitary* if S is an invertible isometry.

As previously noted, every isometry is injective. Every injective operator on a finite-dimensional vector space is invertible (see 3.65). A standing assumption for this chapter is that V is a finite-dimensional inner product space. Thus we could delete the word “invertible” from the definition above without changing the meaning. The unnecessary word “invertible” has been retained in the definition above for consistency with the definition readers may encounter when learning about inner product spaces that are not necessarily finite-dimensional.

Although the words “unitary” and “isometry” mean the same thing for operators on finite-dimensional inner product spaces, remember that a unitary operator maps a vector space to itself, while an isometry maps a vector space to another (possibly different) vector space.

7.52 example: rotation of $\mathbf{R}^2$

Suppose $\theta \in \mathbf{R}$ and S is the operator on $\mathbf{F}^2$ whose matrix with respect to the standard basis of $\mathbf{F}^2$ is

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The two columns of this matrix form an orthonormal list in $\mathbf{F}^2$; hence S is an isometry [by the equivalence of (a) and (e) in 7.49]. Thus S is a unitary operator.

If $\mathbf{F} = \mathbf{R}$, then S is the operator of counterclockwise rotation by θ radians around the origin of $\mathbf{R}^2$. This observation gives us another way to think about why S is an isometry, because each rotation around the origin of $\mathbf{R}^2$ preserves norms.

The next result (7.53) lists several conditions that are equivalent to being a unitary operator. All the conditions equivalent to being an isometry in 7.49 should be added to this list. The extra conditions in 7.53 arise because of limiting the context to linear maps from a vector space to itself. For example, 7.49 shows that a linear map $S \in \mathcal{L}(V, W)$ is an isometry if and only if $S^*S = I$, while 7.53 shows that an operator $S \in \mathcal{L}(V)$ is a unitary operator if and only if $S^*S = SS^* = I$.

Another difference is that 7.49(d) mentions an orthonormal list, while 7.53(d) mentions an orthonormal basis. Also, 7.49(e) mentions the columns of $\mathcal{M}(T)$, while 7.53(e) mentions the rows of $\mathcal{M}(T)$. Furthermore, $\mathcal{M}(T)$ in 7.49(e) is with respect to an orthonormal basis of V and an orthonormal basis of W , while $\mathcal{M}(T)$ in 7.53(e) is with respect to a single basis of V doing double duty.

7.53 characterizations of unitary operators

Suppose $S \in \mathcal{L}(V)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V . Then the following are equivalent.

- (a) S is a unitary operator.
- (b) $S^*S = SS^* = I$.
- (c) S is invertible and $S^{-1} = S^*$.
- (d) $Se_1, \dots, Se_n$ is an orthonormal basis of V .
- (e) The rows of $\mathcal{M}(S, (e_1, \dots, e_n))$ form an orthonormal basis of $\mathbf{F}^n$ with respect to the Euclidean inner product.
- (f) S^* is a unitary operator.

Proof First suppose (a) holds, so S is a unitary operator. Hence

$$S^*S = I$$

by the equivalence of (a) and (b) in 7.49. Multiply both sides of this equation by S^{-1} on the right, getting $S^* = S^{-1}$. Thus $SS^* = SS^{-1} = I$, as desired, proving that (a) implies (b).

The definitions of invertible and inverse show that (b) implies (c).

Now suppose (c) holds, so S is invertible and $S^{-1} = S^*$. Thus $S^*S = I$. Hence $Se_1, \dots, Se_n$ is an orthonormal list in V , by the equivalence of (b) and (d) in 7.49. The length of this list equals $\dim V$. Thus $Se_1, \dots, Se_n$ is an orthonormal basis of V , proving that (c) implies (d).

Now suppose (d) holds, so $Se_1, \dots, Se_n$ is an orthonormal basis of V . The equivalence of (a) and (d) in 7.49 shows that S is a unitary operator. Thus

$$(S^*)^*S^* = SS^* = I,$$

where the last equation holds because we already showed that (a) implies (b) in this result. The equation above and the equivalence of (a) and (b) in 7.49 show that S^* is an isometry. Thus the columns of $\mathcal{M}(S^*, (e_1, \dots, e_n))$ form an orthonormal basis of $\mathbf{F}^n$ [by the equivalence of (a) and (e) of 7.49]. The rows of $\mathcal{M}(S, (e_1, \dots, e_n))$ are the complex conjugates of the columns of $\mathcal{M}(S^*, (e_1, \dots, e_n))$. Thus the rows of $\mathcal{M}(S, (e_1, \dots, e_n))$ form an orthonormal basis of $\mathbf{F}^n$, proving that (d) implies (e).

Now suppose (e) holds. Thus the columns of $\mathcal{M}(S^*, (e_1, \dots, e_n))$ form an orthonormal basis of $\mathbf{F}^n$. The equivalence of (a) and (e) in 7.49 shows that S^* is an isometry, proving that (e) implies (f).

Now suppose (f) holds, so S^* is a unitary operator. The chain of implications we have already proved in this result shows that (a) implies (f). Applying this result to S^* shows that $(S^*)^*$ is a unitary operator, proving that (f) implies (a).

We have shown that (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d) $\Rightarrow$ (e) $\Rightarrow$ (f) $\Rightarrow$ (a), completing the proof. ■

Recall our analogy between $\mathbf{C}$ and $\mathcal{L}(V)$. Under this analogy, a complex number z corresponds to an operator $S \in \mathcal{L}(V)$, and $\bar{z}$ corresponds to S^* . The real numbers ($z = \bar{z}$) correspond to the self-adjoint operators ($S = S^*$), and the nonnegative numbers correspond to the (badly named) positive operators.

Another distinguished subset of $\mathbf{C}$ is the unit circle, which consists of the complex numbers z such that $|z| = 1$. The condition $|z| = 1$ is equivalent to the condition $\bar{z}z = 1$. Under our analogy, this corresponds to the condition $S^*S = I$, which is equivalent to S being a unitary operator. Hence the analogy shows that the unit circle in $\mathbf{C}$ corresponds to the set of unitary operators. In the next two results, this analogy appears in the eigenvalues of unitary operators. Also see Exercise 15 for another example of this analogy.

7.54 eigenvalues of unitary operators have absolute value 1

Suppose λ is an eigenvalue of a unitary operator. Then $|\lambda| = 1$.

Proof Suppose $S \in \mathcal{L}(V)$ is a unitary operator and λ is an eigenvalue of S . Let $v \in V$ be such that $v \neq 0$ and $Sv = \lambda v$. Then

$$|\lambda| \|v\| = \|\lambda v\| = \|Sv\| = \|v\|.$$

Thus $|\lambda| = 1$, as desired. ■

The next result characterizes unitary operators on finite-dimensional complex inner product spaces, using the complex spectral theorem as the main tool.

7.55 description of unitary operators on complex inner product spaces

Suppose $\mathbf{F} = \mathbf{C}$ and $S \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) S is a unitary operator.
- (b) There is an orthonormal basis of V consisting of eigenvectors of S whose corresponding eigenvalues all have absolute value 1.

Proof Suppose (a) holds, so S is a unitary operator. The equivalence of (a) and (b) in 7.53 shows that S is normal. Thus the complex spectral theorem (7.31) shows that there is an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of S . Every eigenvalue of S has absolute value 1 (by 7.54), completing the proof that (a) implies (b).

Now suppose (b) holds. Let $e_1, \dots, e_n$ be an orthonormal basis of V consisting of eigenvectors of S whose corresponding eigenvalues $\lambda_1, \dots, \lambda_n$ all have absolute value 1. Then $Se_1, \dots, Se_n$ is also an orthonormal basis of V because

$$\langle Se_j, Se_k \rangle = \langle \lambda_j e_j, \lambda_k e_k \rangle = \lambda_j \overline{\lambda_k} \langle e_j, e_k \rangle = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k \end{cases}$$

for all $j, k = 1, \dots, n$. Thus the equivalence of (a) and (d) in 7.53 shows that S is unitary, proving that (b) implies (a). ■

QR Factorization

In this subsection, we shift our attention from operators to matrices. This switch should give you good practice in identifying an operator with a square matrix (after picking a basis of the vector space on which the operator is defined). You should also become more comfortable with translating concepts and results back and forth between the context of operators and the context of square matrices.

When starting with n -by- n matrices instead of operators, unless otherwise specified assume that the associated operators live on $\mathbf{F}^n$ (with the Euclidean inner product) and that their matrices are computed with respect to the standard basis of $\mathbf{F}^n$.

We begin by making the following definition, transferring the notion of a unitary operator to a unitary matrix.

7.56 definition: *unitary matrix*

An n -by- n matrix is called *unitary* if its columns form an orthonormal list in $\mathbf{F}^n$.

In the definition above, we could have replaced “orthonormal list in $\mathbf{F}^n$ ” with “orthonormal basis of $\mathbf{F}^n$ ” because every orthonormal list of length n in an n -dimensional inner product space is an orthonormal basis. If $S \in \mathcal{L}(V)$ and $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are orthonormal bases of V , then S is a unitary operator if and only if $\mathcal{M}(S, (e_1, \dots, e_n), (f_1, \dots, f_n))$ is a unitary matrix, as shown by the equivalence of (a) and (e) in 7.49. Also note that we could also have replaced “columns” in the definition above with “rows” by using the equivalence between conditions (a) and (e) in 7.53.

The next result, whose proof will be left as an exercise for the reader, gives some equivalent conditions for a square matrix to be unitary. In (c), Qv denotes the matrix product of Q and v , identifying elements of $\mathbf{F}^n$ with n -by-1 matrices (sometimes called column vectors). The norm in (c) below is the usual Euclidean norm on $\mathbf{F}^n$ that comes from the Euclidean inner product. In (d), Q^* denotes the conjugate transpose of the matrix Q , which corresponds to the adjoint of the associated operator.

7.57 characterizations of unitary matrices

Suppose Q is an n -by- n matrix. Then the following are equivalent.

- (a) Q is a unitary matrix.
- (b) The rows of Q form an orthonormal list in $\mathbf{F}^n$.
- (c) $\|Qv\| = \|v\|$ for every $v \in \mathbf{F}^n$.
- (d) $Q^*Q = QQ^* = I$, the n -by- n matrix with 1's on the diagonal and 0's elsewhere.

The QR factorization stated and proved below is the main tool in the widely used QR algorithm (not discussed here) for finding good approximations to eigenvalues and eigenvectors of square matrices. In the result below, if the matrix A is in $\mathbf{F}^{n,n}$, then the matrices Q and R are also in $\mathbf{F}^{n,n}$.

7.58 QR factorization

Suppose A is a square matrix with linearly independent columns. Then there exist unique matrices Q and R such that Q is unitary, R is upper triangular with only positive numbers on its diagonal, and

$$A = QR.$$

Proof Let $v_1, \dots, v_n$ denote the columns of A , thought of as elements of $\mathbf{F}^n$. Apply the Gram–Schmidt procedure (6.32) to the list $v_1, \dots, v_n$, getting an orthonormal basis $e_1, \dots, e_n$ of $\mathbf{F}^n$ such that

$$7.59 \quad \text{span}(v_1, \dots, v_k) = \text{span}(e_1, \dots, e_k)$$

for each $k = 1, \dots, n$. Let R be the n -by- n matrix defined by

$$R_{j,k} = \langle v_k, e_j \rangle,$$

where $R_{j,k}$ denotes the entry in row j , column k of R . If $j > k$, then e_j is orthogonal to $\text{span}(e_1, \dots, e_k)$ and hence e_j is orthogonal to v_k (by 7.59). In other words, if $j > k$ then $\langle v_k, e_j \rangle = 0$. Thus R is an upper-triangular matrix.

Let Q be the unitary matrix whose columns are $e_1, \dots, e_n$. If $k \in \{1, \dots, n\}$, then the k^{th} column of QR equals a linear combination of the columns of Q , with the coefficients for the linear combination coming from the k^{th} column of R —see 3.51(a). Hence the k^{th} column of QR equals

$$\langle v_k, e_1 \rangle e_1 + \dots + \langle v_k, e_k \rangle e_k,$$

which equals v_k [by 6.30(a)], the k^{th} column of A . Thus $A = QR$, as desired.

The equations defining the Gram–Schmidt procedure (see 6.32) show that each v_k equals a positive multiple of e_k plus a linear combination of $e_1, \dots, e_{k-1}$. Thus each $\langle v_k, e_k \rangle$ is a positive number. Hence all entries on the diagonal of R are positive numbers, as desired.

Finally, to show that Q and R are unique, suppose we also have $A = \widehat{Q}\widehat{R}$, where $\widehat{Q}$ is unitary and $\widehat{R}$ is upper triangular with only positive numbers on its diagonal. Let $q_1, \dots, q_n$ denote the columns of $\widehat{Q}$. Thinking of matrix multiplication as above, we see that each v_k is a linear combination of $q_1, \dots, q_k$, with the coefficients coming from the k^{th} column of $\widehat{R}$. This implies that $\text{span}(v_1, \dots, v_k) = \text{span}(q_1, \dots, q_k)$ and $\langle v_k, q_k \rangle > 0$. The uniqueness of the orthonormal lists satisfying these conditions (see Exercise 10 in Section 6B) now shows that $q_k = e_k$ for each $k = 1, \dots, n$. Hence $\widehat{Q} = Q$, which then implies that $\widehat{R} = R$, completing the proof of uniqueness. ■

The proof of the QR factorization shows that the columns of the unitary matrix can be computed by applying the Gram–Schmidt procedure to the columns of the matrix to be factored. The next example illustrates the computation of the QR factorization based on the proof that we just completed.

7.60 example: *QR factorization of a 3-by-3 matrix*

To find the QR factorization of the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -4 \\ 0 & 3 & 2 \end{pmatrix},$$

follow the proof of 7.58. Thus set v_1, v_2, v_3 equal to the columns of A :

$$v_1 = (1, 0, 0), \quad v_2 = (2, 1, 3), \quad v_3 = (1, -4, 2).$$

Apply the Gram–Schmidt procedure to v_1, v_2, v_3 , producing the orthonormal list

$$e_1 = (1, 0, 0), \quad e_2 = \left(0, \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right), \quad e_3 = \left(0, -\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}\right).$$

Still following the proof of 7.58, let Q be the unitary matrix whose columns are e_1, e_2, e_3 :

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ 0 & \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{pmatrix}.$$

As in the proof of 7.58, let R be the 3-by-3 matrix whose entry in row j , column k is $\langle v_k, e_j \rangle$, which gives

$$R = \begin{pmatrix} 1 & 2 & 1 \\ 0 & \sqrt{10} & \frac{\sqrt{10}}{5} \\ 0 & 0 & \frac{7\sqrt{10}}{5} \end{pmatrix}.$$

Note that R is indeed an upper-triangular matrix with only positive numbers on the diagonal, as required by the QR factorization.

Now matrix multiplication can verify that $A = QR$ is the desired factorization of A :

$$QR = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ 0 & \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 0 & \sqrt{10} & \frac{\sqrt{10}}{5} \\ 0 & 0 & \frac{7\sqrt{10}}{5} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -4 \\ 0 & 3 & 2 \end{pmatrix} = A.$$

Thus $A = QR$, as expected.

The QR factorization will be the major tool used in the proof of the Cholesky factorization (7.63) in the next subsection. For another nice application of the QR factorization, see the proof of Hadamard's inequality (9.66).

If a QR factorization is available, then it can be used to solve a corresponding system of linear equations without using Gaussian elimination. Specifically, suppose A is an n -by- n square matrix with linearly independent columns. Suppose that $b \in \mathbf{F}^n$ and we want to solve the equation $Ax = b$ for $x = (x_1, \dots, x_n) \in \mathbf{F}^n$ (as usual, we are identifying elements of $\mathbf{F}^n$ with n -by-1 column vectors).

Suppose $A = QR$, where Q is unitary and R is upper triangular with only positive numbers on its diagonal (Q and R are computable from A using just the Gram–Schmidt procedure, as shown in the proof of 7.58). The equation $Ax = b$ is equivalent to the equation $QRx = b$. Multiplying both sides of this last equation by Q^* on the left and using 7.57(d) gives the equation

$$Rx = Q^*b.$$

The matrix Q^* is the conjugate transpose of the matrix Q . Thus computing Q^*b is straightforward. Because R is an upper-triangular matrix with positive numbers on its diagonal, the system of linear equations represented by the equation above can quickly be solved by first solving for x_n , then for x_{n-1} , and so on.

Cholesky Factorization

We begin this subsection with a characterization of positive invertible operators in terms of inner products.

7.61 positive invertible operator

A self-adjoint operator $T \in \mathcal{L}(V)$ is a positive invertible operator if and only if $\langle Tv, v \rangle > 0$ for every nonzero $v \in V$.

Proof First suppose T is a positive invertible operator. If $v \in V$ and $v \neq 0$, then because T is invertible we have $Tv \neq 0$. This implies that $\langle Tv, v \rangle \neq 0$ (by 7.43). Hence $\langle Tv, v \rangle > 0$.

To prove the implication in the other direction, suppose now that $\langle Tv, v \rangle > 0$ for every nonzero $v \in V$. Thus $Tv \neq 0$ for every nonzero $v \in V$. Hence T is injective. Thus T is invertible, as desired. ■

The next definition transfers the result above to the language of matrices. Here we are using the usual Euclidean inner product on $\mathbf{F}^n$ and identifying elements of $\mathbf{F}^n$ with n -by-1 column vectors.

7.62 definition: positive definite

A matrix $B \in \mathbf{F}^{n,n}$ is called *positive definite* if $B^* = B$ and

$$\langle Bx, x \rangle > 0$$

for every nonzero $x \in \mathbf{F}^n$.

A matrix is upper triangular if and only if its conjugate transpose is lower triangular (meaning that all entries above the diagonal are 0). The factorization below, which has important consequences in computational linear algebra, writes a positive definite matrix as the product of a lower triangular matrix and its conjugate transpose.

Our next result is solely about matrices, although the proof makes use of the identification of results about operators with results about square matrices. In the result below, if the matrix B is in $\mathbf{F}^{n,n}$, then the matrix R is also in $\mathbf{F}^{n,n}$.

7.63 Cholesky factorization

Suppose B is a positive definite matrix. Then there exists a unique upper-triangular matrix R with only positive numbers on its diagonal such that

$$B = R^*R.$$

Proof Because B is positive definite, there exists an invertible square matrix A of the same size as B such that $B = A^*A$ [by the equivalence of (a) and (f) in 7.38].

Let $A = QR$ be the QR factorization of A (see 7.58), where Q is unitary and R is upper triangular with only positive numbers on its diagonal. Then $A^* = R^*Q^*$.

Thus

$$B = A^*A = R^*Q^*QR = R^*R,$$

André-Louis Cholesky (1875–1918) discovered this factorization, which was published posthumously in 1924.

as desired.

To prove the uniqueness part of this result, suppose S is an upper-triangular matrix with only positive numbers on its diagonal and $B = S^*S$. The matrix S is invertible because B is invertible (see Exercise 11 in Section 3D). Multiplying both sides of the equation $B = S^*S$ by S^{-1} on the right gives the equation $BS^{-1} = S^*$.

Let A be the matrix from the first paragraph of this proof. Then

$$\begin{aligned}(AS^{-1})^*(AS^{-1}) &= (S^*)^{-1}A^*AS^{-1} \\ &= (S^*)^{-1}BS^{-1} \\ &= (S^*)^{-1}S^* \\ &= I.\end{aligned}$$

Thus AS^{-1} is unitary.

Hence $A = (AS^{-1})S$ is a factorization of A as the product of a unitary matrix and an upper-triangular matrix with only positive numbers on its diagonal. The uniqueness of the QR factorization, as stated in 7.58, now implies that $S = R$. ■

In the first paragraph of the proof above, we could have chosen A to be the unique positive definite matrix that is a square root of B (see 7.39). However, the proof was presented with the more general choice of A because for specific positive definite matrices B , it may be easier to find a different choice of A .

Exercises 7D

- 1 Suppose $\dim V \geq 2$ and $S \in \mathcal{L}(V, W)$. Prove that S is an isometry if and only if Se_1, Se_2 is an orthonormal list in W for every orthonormal list e_1, e_2 of length two in V .
- 2 Suppose $T \in \mathcal{L}(V, W)$ and $T \neq 0$. Prove that T is a scalar multiple of an isometry if and only if T preserves orthogonality.
The phrase “ T preserves orthogonality” means that $\langle Tu, Tv \rangle = 0$ for all $u, v \in V$ such that $\langle u, v \rangle = 0$.
- 3 (a) Show that the product of two unitary operators on V is a unitary operator.
 (b) Show that the inverse of a unitary operator on V is a unitary operator.
This exercise shows that the set of unitary operators on V is a group, where the group operation is the usual product of two operators.
- 4 Suppose $\mathbf{F} = \mathbf{C}$ and $A, B \in \mathcal{L}(V)$ are self-adjoint. Show that $A + iB$ is unitary if and only if $AB = BA$ and $A^2 + B^2 = I$.
- 5 Suppose $S \in \mathcal{L}(V)$. Prove that the following are equivalent.
 (a) S is a self-adjoint unitary operator.
 (b) $S = 2P - I$ for some orthogonal projection P on V .
 (c) There exists a subspace U of V such that $Su = u$ for every $u \in U$ and $Sw = -w$ for every $w \in U^\perp$.
- 6 Suppose T_1, T_2 are both normal operators on $\mathbf{F}^3$ with 2, 5, 7 as eigenvalues. Prove that there exists a unitary operator $S \in \mathcal{L}(\mathbf{F}^3)$ such that $T_1 = S^*T_2S$.
- 7 Give an example of two self-adjoint operators $T_1, T_2 \in \mathcal{L}(\mathbf{F}^4)$ such that the eigenvalues of both operators are 2, 5, 7 but there does not exist a unitary operator $S \in \mathcal{L}(\mathbf{F}^4)$ such that $T_1 = S^*T_2S$. Be sure to explain why there is no unitary operator with the required property.
- 8 Prove or give a counterexample: If $S \in \mathcal{L}(V)$ and there exists an orthonormal basis $e_1, \dots, e_n$ of V such that $\|Se_k\| = 1$ for each e_k , then S is a unitary operator.
- 9 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Suppose every eigenvalue of T has absolute value 1 and $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that T is a unitary operator.
- 10 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is a self-adjoint operator such that $\|Tv\| \leq \|v\|$ for all $v \in V$.
 (a) Show that $I - T^2$ is a positive operator.
 (b) Show that $T + i\sqrt{I - T^2}$ is a unitary operator.
- 11 Suppose $S \in \mathcal{L}(V)$. Prove that S is a unitary operator if and only if

$$\{Sv : v \in V \text{ and } \|v\| \leq 1\} = \{v \in V : \|v\| \leq 1\}.$$
- 12 Prove or give a counterexample: If $S \in \mathcal{L}(V)$ is invertible and $\|S^{-1}v\| = \|Sv\|$ for every $v \in V$, then S is unitary.

- 13 Explain why the columns of a square matrix of complex numbers form an orthonormal list in $\mathbf{C}^n$ if and only if the rows of the matrix form an orthonormal list in $\mathbf{C}^n$.
- 14 Suppose $v \in V$ with $\|v\| = 1$ and $b \in \mathbf{F}$. Also suppose $\dim V \geq 2$. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $\langle Sv, v \rangle = b$ if and only if $|b| \leq 1$.
- 15 Suppose T is a unitary operator on V such that $T - I$ is invertible.
- Prove that $(T + I)(T - I)^{-1}$ is a skew operator (meaning that it equals the negative of its adjoint).
 - Prove that if $\mathbf{F} = \mathbf{C}$, then $i(T + I)(T - I)^{-1}$ is a self-adjoint operator.

The function $z \mapsto i(z + 1)(z - 1)^{-1}$ maps the unit circle in $\mathbf{C}$ (except for the point 1) to $\mathbf{R}$. Thus (b) illustrates the analogy between the unitary operators and the unit circle in $\mathbf{C}$, along with the analogy between the self-adjoint operators and $\mathbf{R}$.

- 16 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is self-adjoint. Prove that $(T + iI)(T - iI)^{-1}$ is a unitary operator and 1 is not an eigenvalue of this operator.
- 17 Explain why the characterizations of unitary matrices given by 7.57 hold.
- 18 A square matrix A is called *symmetric* if it equals its transpose. Prove that if A is a symmetric matrix with real entries, then there exists a unitary matrix Q with real entries such that Q^*AQ is a diagonal matrix.
- 19 Suppose n is a positive integer. For this exercise, we adopt the notation that a typical element z of $\mathbf{C}^n$ is denoted by $z = (z_0, z_1, \dots, z_{n-1})$. Define linear functionals $\omega_0, \omega_1, \dots, \omega_{n-1}$ on $\mathbf{C}^n$ by

$$\omega_j(z_0, z_1, \dots, z_{n-1}) = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} z_m e^{-2\pi i j m / n}.$$

The *discrete Fourier transform* is the operator $\mathcal{F} : \mathbf{C}^n \rightarrow \mathbf{C}^n$ defined by

$$\mathcal{F}z = (\omega_0(z), \omega_1(z), \dots, \omega_{n-1}(z)).$$

- Show that $\mathcal{F}$ is a unitary operator on $\mathbf{C}^n$.
- Show that if $(z_0, \dots, z_{n-1}) \in \mathbf{C}^n$ and z_n is defined to equal z_0 , then

$$\mathcal{F}^{-1}(z_0, z_1, \dots, z_{n-1}) = \mathcal{F}(z_n, z_{n-1}, \dots, z_1).$$

- Show that $\mathcal{F}^4 = I$.

The discrete Fourier transform has many important applications in data analysis. The usual Fourier transform involves expressions of the form $\int_{-\infty}^{\infty} f(x)e^{-2\pi i t x} dx$ for complex-valued integrable functions f defined on $\mathbf{R}$.

- 20 Suppose A is a square matrix with linearly independent columns. Prove that there exist unique matrices R and Q such that R is lower triangular with only positive numbers on its diagonal, Q is unitary, and $A = RQ$.

7E Singular Value Decomposition

Singular Values

We will need the following result in this section.

7.64 properties of T^*T

Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) T^*T is a positive operator on V ;
- (b) $\text{null } T^*T = \text{null } T$;
- (c) $\text{range } T^*T = \text{range } T^*$;
- (d) $\dim \text{range } T = \dim \text{range } T^* = \dim \text{range } T^*T$.

Proof

- (a) We have

$$(T^*T)^* = T^*(T^*)^* = T^*T.$$

Thus T^*T is self-adjoint.

If $v \in V$, then

$$\langle (T^*T)v, v \rangle = \langle T^*(Tv), v \rangle = \langle Tv, Tv \rangle = \|Tv\|^2 \geq 0.$$

Thus T^*T is a positive operator.

- (b) First suppose $v \in \text{null } T^*T$. Then

$$\|Tv\|^2 = \langle Tv, Tv \rangle = \langle T^*Tv, v \rangle = \langle 0, v \rangle = 0.$$

Thus $Tv = 0$, proving that $\text{null } T^*T \subseteq \text{null } T$.

The inclusion in the other direction is clear, because if $v \in V$ and $Tv = 0$, then $T^*Tv = 0$.

Thus $\text{null } T^*T = \text{null } T$, completing the proof of (b).

- (c) We already know from (a) that T^*T is self-adjoint. Thus

$$\text{range } T^*T = (\text{null } T^*T)^\perp = (\text{null } T)^\perp = \text{range } T^*,$$

where the first and last equalities come from 7.6 and the second equality comes from (b).

- (d) To verify the first equation in (d), note that

$$\dim \text{range } T = \dim(\text{null } T^*)^\perp = \dim W - \dim \text{null } T^* = \dim \text{range } T^*,$$

where the first equality comes from 7.6(d), the second equality comes from 6.51, and the last equality comes from the fundamental theorem of linear maps (3.21).

The equality $\dim \text{range } T^* = \dim \text{range } T^*T$ follows from (c). ■

The eigenvalues of an operator tell us something about the behavior of the operator. Another collection of numbers, called the singular values, is also useful. Eigenspaces and the notation E (used in the examples) were defined in 5.52.

7.65 definition: *singular values*

Suppose $T \in \mathcal{L}(V, W)$. The *singular values* of T are the nonnegative square roots of the eigenvalues of T^*T , listed in decreasing order, each included as many times as the dimension of the corresponding eigenspace of T^*T .

7.66 example: *singular values of an operator on $\mathbf{F}^4$*

Define $T \in \mathcal{L}(\mathbf{F}^4)$ by $T(z_1, z_2, z_3, z_4) = (0, 3z_1, 2z_2, -3z_4)$. A calculation shows that

$$T^*T(z_1, z_2, z_3, z_4) = (9z_1, 4z_2, 0, 9z_4),$$

as you should verify. Thus the standard basis of $\mathbf{F}^4$ diagonalizes T^*T , and we see that the eigenvalues of T^*T are 9, 4, and 0. Also, the dimensions of the eigenspaces corresponding to the eigenvalues are

$$\dim E(9, T^*T) = 2 \quad \text{and} \quad \dim E(4, T^*T) = 1 \quad \text{and} \quad \dim E(0, T^*T) = 1.$$

Taking nonnegative square roots of these eigenvalues of T^*T and using dimension information from above, we conclude that the singular values of T are 3, 3, 2, 0.

The only eigenvalues of T are -3 and 0 . Thus in this case, the collection of eigenvalues did not pick up the number 2 that appears in the definition (and hence the behavior) of T , but the list of singular values does include 2.

7.67 example: *singular values of a linear map from $\mathbf{F}^4$ to $\mathbf{F}^3$*

Suppose $T \in \mathcal{L}(\mathbf{F}^4, \mathbf{F}^3)$ has matrix (with respect to the standard bases)

$$\begin{pmatrix} 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

You can verify that the matrix of T^*T is

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 25 \end{pmatrix}$$

and that the eigenvalues of the operator T^*T are 25, 2, 0, with $\dim E(25, T^*T) = 1$, $\dim E(2, T^*T) = 1$, and $\dim E(0, T^*T) = 2$. Thus the singular values of T are $5, \sqrt{2}, 0, 0$.

See Exercise 2 for a characterization of the positive singular values.

7.68 role of positive singular values

Suppose that $T \in \mathcal{L}(V, W)$. Then

- (a) T is injective $\iff 0$ is not a singular value of T ;
- (b) the number of positive singular values of T equals $\dim \text{range } T$;
- (c) T is surjective $\iff$ number of positive singular values of T equals $\dim W$.

Proof The linear map T is injective if and only if $\text{null } T = \{0\}$, which happens if and only if $\text{null } T^*T = \{0\}$ [by 7.64(b)], which happens if and only if 0 is not an eigenvalue of T^*T , which happens if and only if 0 is not a singular value of T , completing the proof of (a).

The spectral theorem applied to T^*T shows that $\dim \text{range } T^*T$ equals the number of positive eigenvalues of T^*T (counting repetitions). Thus 7.64(d) implies that $\dim \text{range } T$ equals the number of positive singular values of T , proving (b).

Use (b) and 2.39 to show that (c) holds. ■

The table below compares eigenvalues with singular values.

list of eigenvalues	list of singular values
context: vector spaces	context: inner product spaces
defined only for linear maps from a vector space to itself	defined for linear maps from an inner product space to a possibly different inner product space
can be arbitrary real numbers (if $\mathbf{F} = \mathbf{R}$) or complex numbers (if $\mathbf{F} = \mathbf{C}$)	are nonnegative numbers
can be the empty list if $\mathbf{F} = \mathbf{R}$	length of list equals dimension of domain
includes $0 \iff$ operator is not invertible	includes $0 \iff$ linear map is not injective
no standard order, especially if $\mathbf{F} = \mathbf{C}$	always listed in decreasing order

The next result nicely characterizes isometries in terms of singular values.

7.69 isometries characterized by having all singular values equal 1

Suppose that $S \in \mathcal{L}(V, W)$. Then

$$S \text{ is an isometry} \iff \text{all singular values of } S \text{ equal } 1.$$

Proof We have

$$\begin{aligned}
 S \text{ is an isometry} &\iff S^*S = I \\
 &\iff \text{all eigenvalues of } S^*S \text{ equal } 1 \\
 &\iff \text{all singular values of } S \text{ equal } 1,
 \end{aligned}$$

where the first equivalence comes from 7.49 and the second equivalence comes from the spectral theorem (7.29 or 7.31) applied to the self-adjoint operator S^*S . ■

SVD for Linear Maps and for Matrices

The next result shows that every linear map from V to W has a remarkably clean description in terms of its singular values and orthonormal lists in V and W . In the next section we will see several important applications of the singular value decomposition (often called the SVD).

*The singular value decomposition is useful in computational linear algebra because good techniques exist for approximating eigenvalues and eigenvectors of positive operators such as T^*T , whose eigenvalues and eigenvectors lead to the singular value decomposition.*

7.70 singular value decomposition

Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of T are $s_1, \dots, s_m$. Then there exist orthonormal lists $e_1, \dots, e_m$ in V and $f_1, \dots, f_m$ in W such that

$$7.71 \quad Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m$$

for every $v \in V$.

Proof Let $s_1, \dots, s_n$ denote the singular values of T (thus $n = \dim V$). Because T^*T is a positive operator [see 7.64(a)], the spectral theorem implies that there exists an orthonormal basis $e_1, \dots, e_n$ of V with

$$7.72 \quad T^*Te_k = s_k^2 e_k$$

for each $k = 1, \dots, n$.

For each $k = 1, \dots, m$, let

$$7.73 \quad f_k = \frac{Te_k}{s_k}.$$

If $j, k \in \{1, \dots, m\}$, then

$$\langle f_j, f_k \rangle = \frac{1}{s_j s_k} \langle Te_j, Te_k \rangle = \frac{1}{s_j s_k} \langle e_j, T^*Te_k \rangle = \frac{s_k}{s_j} \langle e_j, e_k \rangle = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k. \end{cases}$$

Thus $f_1, \dots, f_m$ is an orthonormal list in W .

If $k \in \{1, \dots, n\}$ and $k > m$, then $s_k = 0$ and hence $T^*Te_k = 0$ (by 7.72), which implies that $Te_k = 0$ [by 7.64(b)].

Suppose $v \in V$. Then

$$\begin{aligned} Tv &= T(\langle v, e_1 \rangle e_1 + \cdots + \langle v, e_n \rangle e_n) \\ &= \langle v, e_1 \rangle Te_1 + \cdots + \langle v, e_m \rangle Te_m \\ &= s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m, \end{aligned}$$

where the last index in the first line switched from n to m in the second line because $Te_k = 0$ if $k > m$ (as noted in the paragraph above) and the third line follows from 7.73. The equation above is our desired result. ■

Suppose $T \in \mathcal{L}(V, W)$, the positive singular values of T are $s_1, \dots, s_m$, and $e_1, \dots, e_m$ and $f_1, \dots, f_m$ are as in the singular value decomposition 7.70. The orthonormal list $e_1, \dots, e_m$ can be extended to an orthonormal basis $e_1, \dots, e_{\dim V}$ of V and the orthonormal list $f_1, \dots, f_m$ can be extended to an orthonormal basis $f_1, \dots, f_{\dim W}$ of W . The formula 7.71 shows that

$$Te_k = \begin{cases} s_k f_k & \text{if } 1 \leq k \leq m, \\ 0 & \text{if } m < k \leq \dim V. \end{cases}$$

Thus the matrix of T with respect to the orthonormal bases $(e_1, \dots, e_{\dim V})$ and $(f_1, \dots, f_{\dim W})$ has the simple form

$$\mathcal{M}(T, (e_1, \dots, e_{\dim V}), (f_1, \dots, f_{\dim W}))_{j,k} = \begin{cases} s_k & \text{if } 1 \leq j = k \leq m, \\ 0 & \text{otherwise.} \end{cases}$$

If $\dim V = \dim W$ (as happens, for example, if $W = V$), then the matrix described in the paragraph above is a diagonal matrix. If we extend the definition of diagonal matrix as follows to apply to matrices that are not necessarily square, then we have proved the wonderful result that every linear map from V to W has a diagonal matrix with respect to appropriate orthonormal bases.

7.74 definition: *diagonal matrix*

An M -by- N matrix A is called a *diagonal matrix* if all entries of the matrix are 0 except possibly $A_{k,k}$ for $k = 1, \dots, \min\{M, N\}$.

The table below compares the spectral theorem (7.29 and 7.31) with the singular value decomposition (7.70).

spectral theorem	singular value decomposition
describes only self-adjoint operators (when $\mathbf{F} = \mathbf{R}$) or normal operators (when $\mathbf{F} = \mathbf{C}$)	describes arbitrary linear maps from an inner product space to a possibly different inner product space
produces a single orthonormal basis	produces two orthonormal lists, one for domain space and one for range space, that are not necessarily the same even when range space equals domain space
different proofs depending on whether $\mathbf{F} = \mathbf{R}$ or $\mathbf{F} = \mathbf{C}$	same proof works regardless of whether $\mathbf{F} = \mathbf{R}$ or $\mathbf{F} = \mathbf{C}$

The singular value decomposition gives us a new way to understand the adjoint and the inverse of a linear map. Specifically, the next result shows that given a singular value decomposition of a linear map $T \in \mathcal{L}(V, W)$, we can obtain the adjoint of T simply by interchanging the roles of the e 's and the f 's (see 7.77). Similarly, we can obtain the pseudoinverse $T^\dagger$ (see 6.68) of T by interchanging the roles of the e 's and the f 's and replacing each positive singular value s_k of T with $1/s_k$ (see 7.78).

Recall that the pseudoinverse $T^\dagger$ in 7.78 below equals the inverse T^{-1} if T is invertible [see 6.69(a)].

7.75 singular value decomposition of adjoint and pseudoinverse

Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of T are $s_1, \dots, s_m$. Suppose $e_1, \dots, e_m$ and $f_1, \dots, f_m$ are orthonormal lists in V and W such that

$$7.76 \quad Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m$$

for every $v \in V$. Then

$$7.77 \quad T^*w = s_1 \langle w, f_1 \rangle e_1 + \dots + s_m \langle w, f_m \rangle e_m$$

and

$$7.78 \quad T^\dagger w = \frac{\langle w, f_1 \rangle}{s_1} e_1 + \dots + \frac{\langle w, f_m \rangle}{s_m} e_m$$

for every $w \in W$.

Proof If $v \in V$ and $w \in W$ then

$$\begin{aligned} \langle Tv, w \rangle &= \langle s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m, w \rangle \\ &= s_1 \langle v, e_1 \rangle \langle f_1, w \rangle + \dots + s_m \langle v, e_m \rangle \langle f_m, w \rangle \\ &= \langle v, s_1 \langle w, f_1 \rangle e_1 + \dots + s_m \langle w, f_m \rangle e_m \rangle. \end{aligned}$$

This implies that

$$T^*w = s_1 \langle w, f_1 \rangle e_1 + \dots + s_m \langle w, f_m \rangle e_m,$$

proving 7.77.

To prove 7.78, suppose $w \in W$. Let

$$v = \frac{\langle w, f_1 \rangle}{s_1} e_1 + \dots + \frac{\langle w, f_m \rangle}{s_m} e_m.$$

Apply T to both sides of the equation above, getting

$$\begin{aligned} Tv &= \frac{\langle w, f_1 \rangle}{s_1} Te_1 + \dots + \frac{\langle w, f_m \rangle}{s_m} Te_m \\ &= \langle w, f_1 \rangle f_1 + \dots + \langle w, f_m \rangle f_m \\ &= P_{\text{range } T} w, \end{aligned}$$

where the second line holds because 7.76 implies that $Te_k = s_k f_k$ if $k = 1, \dots, m$, and the last line above holds because 7.76 implies that $f_1, \dots, f_m$ spans $\text{range } T$ and thus is an orthonormal basis of $\text{range } T$ [and hence 6.57(i) applies]. The equation above, the observation that $v \in (\text{null } T)^\perp$ [see Exercise 8(b)], and the definition of $T^\dagger w$ (see 6.68) show that $v = T^\dagger w$, proving 7.78. ■

7.79 example: finding a singular value decomposition

Define $T \in \mathcal{L}(\mathbf{F}^4, \mathbf{F}^3)$ by $T(x_1, x_2, x_3, x_4) = (-5x_4, 0, x_1 + x_2)$. We want to find a singular value decomposition of T . The matrix of T (with respect to the standard bases) is

$$\begin{pmatrix} 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

Thus, as discussed in Example 7.67, the matrix of T^*T is

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 25 \end{pmatrix},$$

and the positive eigenvalues of T^*T are 25, 2, with $\dim E(25, T^*T) = 1$ and $\dim E(2, T^*T) = 1$. Hence the positive singular values of T are 5, $\sqrt{2}$.

Thus to find a singular value decomposition of T , we must find an orthonormal list e_1, e_2 in $\mathbf{F}^4$ and an orthonormal list f_1, f_2 in $\mathbf{F}^3$ such that

$$Tv = 5\langle v, e_1 \rangle f_1 + \sqrt{2}\langle v, e_2 \rangle f_2$$

for all $v \in \mathbf{F}^4$.

An orthonormal basis of $E(25, T^*T)$ is the vector $(0, 0, 0, 1)$; an orthonormal basis of $E(2, T^*T)$ is the vector $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0)$. Thus, following the proof of 7.70, we take

$$e_1 = (0, 0, 0, 1) \quad \text{and} \quad e_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0 \right)$$

and

$$f_1 = \frac{Te_1}{5} = (-1, 0, 0) \quad \text{and} \quad f_2 = \frac{Te_2}{\sqrt{2}} = (0, 0, 1).$$

Then, as expected, we see that e_1, e_2 is an orthonormal list in $\mathbf{F}^4$ and f_1, f_2 is an orthonormal list in $\mathbf{F}^3$ and

$$Tv = 5\langle v, e_1 \rangle f_1 + \sqrt{2}\langle v, e_2 \rangle f_2$$

for all $v \in \mathbf{F}^4$. Thus we have found a singular value decomposition of T .

The next result translates the singular value decomposition from the context of linear maps to the context of matrices. Specifically, the following result gives a factorization of an arbitrary matrix as the product of three nice matrices. The proof gives an explicit construction of these three matrices in terms of the singular value decomposition.

In the next result, the phrase “orthonormal columns” should be interpreted to mean that the columns are orthonormal with respect to the standard Euclidean inner product.

7.80 matrix version of SVD

Suppose A is a p -by- n matrix of rank $m \geq 1$. Then there exist a p -by- m matrix B with orthonormal columns, an m -by- m diagonal matrix D with positive numbers on the diagonal, and an n -by- m matrix C with orthonormal columns such that

$$A = BDC^*.$$

Proof Let $T: \mathbf{F}^n \rightarrow \mathbf{F}^p$ be the linear map whose matrix with respect to the standard bases equals A . Then $\dim \text{range } T = m$ (by 3.78). Let

$$7.81 \quad Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_m \langle v, e_m \rangle f_m$$

be a singular value decomposition of T . Let

$B =$ the p -by- m matrix whose columns are $f_1, \dots, f_m$,

$D =$ the m -by- m diagonal matrix whose diagonal entries are $s_1, \dots, s_m$,

$C =$ the n -by- m matrix whose columns are $e_1, \dots, e_m$.

Let $u_1, \dots, u_m$ denote the standard basis of $\mathbf{F}^m$. If $k \in \{1, \dots, m\}$ then

$$(AC - BD)u_k = Ae_k - B(s_k u_k) = s_k f_k - s_k f_k = 0.$$

Thus $AC = BD$.

Multiply both sides of this last equation by C^* (the conjugate transpose of C) on the right to get

$$ACC^* = BDC^*.$$

Note that the rows of C^* are the complex conjugates of $e_1, \dots, e_m$. Thus if $k \in \{1, \dots, m\}$, then the definition of matrix multiplication shows that $C^*e_k = u_k$; hence $CC^*e_k = e_k$. Thus $ACC^*v = Av$ for all $v \in \text{span}(e_1, \dots, e_m)$.

If $v \in (\text{span}(e_1, \dots, e_m))^\perp$, then $Av = 0$ (as follows from 7.81) and $C^*v = 0$ (as follows from the definition of matrix multiplication). Hence $ACC^*v = Av$ for all $v \in (\text{span}(e_1, \dots, e_m))^\perp$.

Because ACC^* and A agree on $\text{span}(e_1, \dots, e_m)$ and on $(\text{span}(e_1, \dots, e_m))^\perp$, we conclude that $ACC^* = A$. Thus the displayed equation above becomes

$$A = BDC^*,$$

as desired. ■

Note that the matrix A in the result above has pn entries. In comparison, the matrices B , D , and C above have a total of

$$m(p + m + n)$$

entries. Thus if p and n are large numbers and the rank m is considerably less than p and n , then the number of entries that must be stored on a computer to represent A is considerably less than pn .

Exercises 7E

- 1 Suppose $T \in \mathcal{L}(V, W)$. Show that $T = 0$ if and only if all singular values of T are 0.
- 2 Suppose $T \in \mathcal{L}(V, W)$ and $s > 0$. Prove that s is a singular value of T if and only if there exist nonzero vectors $v \in V$ and $w \in W$ such that

$$Tv = sw \quad \text{and} \quad T^*w = sv.$$

The vectors v, w satisfying both equations above are called a **Schmidt pair**.
Erhard Schmidt introduced the concept of singular values in 1907.

- 3 Give an example of $T \in \mathcal{L}(\mathbf{C}^2)$ such that 0 is the only eigenvalue of T and the singular values of T are 5, 0.
- 4 Suppose that $T \in \mathcal{L}(V, W)$, s_1 is the largest singular value of T , and s_n is the smallest singular value of T . Prove that

$$\{\|Tv\| : v \in V \text{ and } \|v\| = 1\} = [s_n, s_1].$$

- 5 Suppose $T \in \mathcal{L}(\mathbf{C}^2)$ is defined by $T(x, y) = (-4y, x)$. Find the singular values of T .
- 6 Find the singular values of the differentiation operator $D \in \mathcal{L}(\mathcal{P}_2(\mathbf{R}))$ defined by $Dp = p'$, where the inner product on $\mathcal{P}_2(\mathbf{R})$ is as in Example 6.34.
- 7 Suppose that $T \in \mathcal{L}(V)$ is self-adjoint or that $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of T , each included in this list as many times as the dimension of the corresponding eigenspace. Show that the singular values of T are $|\lambda_1|, \dots, |\lambda_n|$, after these numbers have been sorted into decreasing order.
- 8 Suppose $T \in \mathcal{L}(V, W)$. Suppose $s_1 \geq s_2 \geq \dots \geq s_m > 0$ and $e_1, \dots, e_m$ is an orthonormal list in V and $f_1, \dots, f_m$ is an orthonormal list in W such that

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m$$

for every $v \in V$.

- (a) Prove that $f_1, \dots, f_m$ is an orthonormal basis of range T .
- (b) Prove that $e_1, \dots, e_m$ is an orthonormal basis of $(\text{null } T)^\perp$.
- (c) Prove that $s_1, \dots, s_m$ are the positive singular values of T .
- (d) Prove that if $k \in \{1, \dots, m\}$, then e_k is an eigenvector of T^*T with corresponding eigenvalue s_k^2 .
- (e) Prove that

$$TT^*w = s_1^2 \langle w, f_1 \rangle f_1 + \dots + s_m^2 \langle w, f_m \rangle f_m$$

for all $w \in W$.

- 9 Suppose $T \in \mathcal{L}(V, W)$. Show that T and T^* have the same positive singular values.
- 10 Suppose $T \in \mathcal{L}(V, W)$ has singular values $s_1, \dots, s_n$. Prove that if T is an invertible linear map, then T^{-1} has singular values

$$\frac{1}{s_n}, \dots, \frac{1}{s_1}.$$

- 11 Suppose that $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_n$ is an orthonormal basis of V . Let $s_1, \dots, s_n$ denote the singular values of T .
- (a) Prove that $\|Tv_1\|^2 + \dots + \|Tv_n\|^2 = s_1^2 + \dots + s_n^2$.
- (b) Prove that if $W = V$ and T is a positive operator, then

$$\langle Tv_1, v_1 \rangle + \dots + \langle Tv_n, v_n \rangle = s_1 + \dots + s_n.$$

See the comment after Exercise 5 in Section 7A.

- 12 (a) Give an example of a finite-dimensional vector space and an operator T on it such that the singular values of T^2 do not equal the squares of the singular values of T .
- (b) Suppose $T \in \mathcal{L}(V)$ is normal. Prove that the singular values of T^2 equal the squares of the singular values of T .
- 13 Suppose $T_1, T_2 \in \mathcal{L}(V)$. Prove that T_1 and T_2 have the same singular values if and only if there exist unitary operators $S_1, S_2 \in \mathcal{L}(V)$ such that $T_1 = S_1 T_2 S_2$.
- 14 Suppose $T \in \mathcal{L}(V, W)$. Let s_n denote the smallest singular value of T . Prove that $s_n \|v\| \leq \|Tv\|$ for every $v \in V$.
- 15 Suppose $T \in \mathcal{L}(V)$ and $s_1 \geq \dots \geq s_n$ are the singular values of T . Prove that if λ is an eigenvalue of T , then $s_1 \geq |\lambda| \geq s_n$.
- 16 Suppose $T \in \mathcal{L}(V, W)$. Prove that $(T^*)^\dagger = (T^\dagger)^*$.
- Compare the result in this exercise to the analogous result for invertible linear maps [see 7.5(f)].*
- 17 Suppose $T \in \mathcal{L}(V)$. Prove that T is self-adjoint if and only if $T^\dagger$ is self-adjoint.

Matrices unfold

Singular values gleam like stars

Order in chaos shines

—written by ChatGPT with input *haiku about SVD*

7F Consequences of Singular Value Decomposition

Norms of Linear Maps

The singular value decomposition leads to the following upper bound for $\|Tv\|$.

7.82 upper bound for $\|Tv\|$

Suppose $T \in \mathcal{L}(V, W)$. Let s_1 be the largest singular value of T . Then

$$\|Tv\| \leq s_1 \|v\|$$

for all $v \in V$.

Proof Let $s_1, \dots, s_m$ denote the positive singular values of T , and let $e_1, \dots, e_m$ be an orthonormal list in V and $f_1, \dots, f_m$ be an orthonormal list in W that provide a singular value decomposition of T . Thus

For a lower bound on $\|Tv\|$, look at Exercise 14 in Section 7E.

$$7.83 \quad Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m$$

for all $v \in V$. Hence if $v \in V$ then

$$\begin{aligned} \|Tv\|^2 &= s_1^2 |\langle v, e_1 \rangle|^2 + \dots + s_m^2 |\langle v, e_m \rangle|^2 \\ &\leq s_1^2 (|\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_m \rangle|^2) \\ &\leq s_1^2 \|v\|^2, \end{aligned}$$

where the last inequality follows from Bessel's inequality (6.26). Taking square roots of both sides of the inequality above shows that $\|Tv\| \leq s_1 \|v\|$, as desired. ■

Suppose $T \in \mathcal{L}(V, W)$ and s_1 is the largest singular value of T . The result above shows that

$$7.84 \quad \|Tv\| \leq s_1 \text{ for all } v \in V \text{ with } \|v\| \leq 1.$$

Taking $v = e_1$ in 7.83 shows that $Te_1 = s_1 f_1$. Because $\|f_1\| = 1$, this implies that $\|Te_1\| = s_1$. Thus because $\|e_1\| = 1$, the inequality in 7.84 leads to the equation

$$7.85 \quad \max\{\|Tv\| : v \in V \text{ and } \|v\| \leq 1\} = s_1.$$

The equation above is the motivation for the following definition, which defines the norm of T to be the left side of the equation above without needing to refer to singular values or the singular value decomposition.

7.86 definition: norm of a linear map, $\|\cdot\|$

Suppose $T \in \mathcal{L}(V, W)$. Then the *norm* of T , denoted by $\|T\|$, is defined by

$$\|T\| = \max\{\|Tv\| : v \in V \text{ and } \|v\| \leq 1\}.$$

In general, the maximum of an infinite set of nonnegative numbers need not exist. However, the discussion before 7.86 shows that the maximum in the definition of the norm of a linear map T from V to W does indeed exist (and equals the largest singular value of T).

We now have two different uses of the word *norm* and the notation $\|\cdot\|$. Our first use of this notation was in connection with an inner product on V , when we defined $\|v\| = \sqrt{\langle v, v \rangle}$ for each $v \in V$. Our second use of the norm notation and terminology is with the definition we just made of $\|T\|$ for $T \in \mathcal{L}(V, W)$. The norm $\|T\|$ for $T \in \mathcal{L}(V, W)$ does not usually come from taking an inner product of T with itself (see Exercise 21). You should be able to tell from the context and from the symbols used which meaning of the norm is intended.

The properties of the norm on $\mathcal{L}(V, W)$ listed below look identical to properties of the norm on an inner product space (see 6.9 and 6.17). The inequality in (d) is called the *triangle inequality*, thus using the same terminology that we used for the norm on V . For the reverse triangle inequality, see Exercise 1.

7.87 basic properties of norms of linear maps

Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $\|T\| \geq 0$;
- (b) $\|T\| = 0 \iff T = 0$;
- (c) $\|\lambda T\| = |\lambda| \|T\|$ for all $\lambda \in \mathbf{F}$;
- (d) $\|S + T\| \leq \|S\| + \|T\|$ for all $S \in \mathcal{L}(V, W)$.

Proof

- (a) Because $\|Tv\| \geq 0$ for every $v \in V$, the definition of $\|T\|$ implies that $\|T\| \geq 0$.
- (b) Suppose $\|T\| = 0$. Thus $Tv = 0$ for all $v \in V$ with $\|v\| \leq 1$. If $u \in V$ with $u \neq 0$, then

$$Tu = \|u\| T\left(\frac{u}{\|u\|}\right) = 0,$$

where the last equality holds because $u/\|u\|$ has norm 1. Because $Tu = 0$ for all $u \in V$, we have $T = 0$.

Conversely, if $T = 0$ then $Tv = 0$ for all $v \in V$ and hence $\|T\| = 0$.

- (c) Suppose $\lambda \in \mathbf{F}$. Then

$$\begin{aligned}\|\lambda T\| &= \max\{\|\lambda Tv\| : v \in V \text{ and } \|v\| \leq 1\} \\ &= |\lambda| \max\{\|Tv\| : v \in V \text{ and } \|v\| \leq 1\} \\ &= |\lambda| \|T\|.\end{aligned}$$

- (d) Suppose $S \in \mathcal{L}(V, W)$. The definition of $\|S + T\|$ implies that there exists $v \in V$ such that $\|v\| \leq 1$ and $\|S + T\| = \|(S + T)v\|$. Now

$$\|S + T\| = \|(S + T)v\| = \|Sv + Tv\| \leq \|Sv\| + \|Tv\| \leq \|S\| + \|T\|,$$

completing the proof of (d). ■

For $S, T \in \mathcal{L}(V, W)$, the quantity $\|S - T\|$ is often called the distance between S and T . Informally, think of the condition that $\|S - T\|$ is a small number as meaning that S and T are close together. For example, Exercise 9 asserts that for every $T \in \mathcal{L}(V)$, there is an invertible operator as close to T as we wish.

7.88 alternative formulas for $\|T\|$

Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $\|T\|$ = the largest singular value of T ;
- (b) $\|T\| = \max\{\|Tv\| : v \in V \text{ and } \|v\| = 1\}$;
- (c) $\|T\|$ = the smallest number c such that $\|Tv\| \leq c\|v\|$ for all $v \in V$.

Proof

(a) See 7.85.

(b) Let $v \in V$ be such that $0 < \|v\| \leq 1$. Let $u = v/\|v\|$. Then

$$\|u\| = \left\| \frac{v}{\|v\|} \right\| = 1 \quad \text{and} \quad \|Tu\| = \left\| T\left(\frac{v}{\|v\|}\right) \right\| = \frac{\|Tv\|}{\|v\|} \geq \|Tv\|.$$

Thus when finding the maximum of $\|Tv\|$ with $\|v\| \leq 1$, we can restrict attention to vectors in V with norm 1, proving (b).

(c) Suppose $v \in V$ and $v \neq 0$. Then the definition of $\|T\|$ implies that

$$\left\| T\left(\frac{v}{\|v\|}\right) \right\| \leq \|T\|,$$

which implies that

$$7.89 \quad \|Tv\| \leq \|T\| \|v\|.$$

Now suppose $c \geq 0$ and $\|Tv\| \leq c\|v\|$ for all $v \in V$. This implies that

$$\|Tv\| \leq c$$

for all $v \in V$ with $\|v\| \leq 1$. Taking the maximum of the left side of the inequality above over all $v \in V$ with $\|v\| \leq 1$ shows that $\|T\| \leq c$. Thus $\|T\|$ is the smallest number c such that $\|Tv\| \leq c\|v\|$ for all $v \in V$. ■

When working with norms of linear maps, you will probably frequently use the inequality 7.89.

For computing an approximation of the norm of a linear map T given the matrix of T with respect to some orthonormal bases, 7.88(a) is likely to be most useful. The matrix of T^*T is quickly computable from matrix multiplication. Then a computer can be asked to find an approximation for the largest eigenvalue of T^*T (excellent numerical algorithms exist for this purpose). Then taking the square root and using 7.88(a) gives an approximation for the norm of T (which usually cannot be computed exactly).

You should verify all assertions in the example below.

7.90 example: *norms*

- If I denotes the usual identity operator on V , then $\|I\| = 1$.
- If $T \in \mathcal{L}(\mathbf{F}^n)$ and the matrix of T with respect to the standard basis of $\mathbf{F}^n$ consists of all 1's, then $\|T\| = n$.
- If $T \in \mathcal{L}(V)$ and V has an orthonormal basis consisting of eigenvectors of T with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$, then $\|T\|$ is the maximum of the numbers $|\lambda_1|, \dots, |\lambda_n|$.
- Suppose $T \in \mathcal{L}(\mathbf{R}^5)$ is the operator whose matrix (with respect to the standard basis) is the 5-by-5 matrix whose entry in row j , column k is $1/(j^2 + k)$. Standard mathematical software shows that the largest singular value of T is approximately 0.8 and the smallest singular value of T is approximately 10^{-6} . Thus $\|T\| \approx 0.8$ and (using Exercise 10 in Section 7E) $\|T^{-1}\| \approx 10^6$. It is not possible to find exact formulas for these norms.

A linear map and its adjoint have the same norm, as shown by the next result.

7.91 *norm of the adjoint*

Suppose $T \in \mathcal{L}(V, W)$. Then $\|T^*\| = \|T\|$.

Proof Suppose $w \in W$. Then

$$\|T^*w\|^2 = \langle T^*w, T^*w \rangle = \langle TT^*w, w \rangle \leq \|TT^*w\| \|w\| \leq \|T\| \|T^*w\| \|w\|.$$

The inequality above implies that

$$\|T^*w\| \leq \|T\| \|w\|,$$

which along with 7.88(c) implies that $\|T^*\| \leq \|T\|$.

Replacing T with T^* in the inequality $\|T^*\| \leq \|T\|$ and then using the equation $(T^*)^* = T$ shows that $\|T\| \leq \|T^*\|$. Thus $\|T^*\| = \|T\|$, as desired. ■

You may want to construct an alternative proof of the result above using Exercise 9 in Section 7E, which asserts that a linear map and its adjoint have the same positive singular values.

Approximation by Linear Maps with Lower-Dimensional Range

The next result is a spectacular application of the singular value decomposition. It says that to best approximate a linear map by a linear map whose range has dimension at most k , chop off the singular value decomposition after the first k terms. Specifically, the linear map T_k in the next result has the property that $\dim \text{range } T_k = k$ and T_k minimizes the distance to T among all linear maps with range of dimension at most k . This result leads to algorithms for compressing huge matrices while preserving their most important information.

7.92 *best approximation by linear map whose range has dimension $\leq k$*

Suppose $T \in \mathcal{L}(V, W)$ and $s_1 \geq \dots \geq s_m$ are the positive singular values of T . Suppose $1 \leq k < m$. Then

$$\min\{\|T - S\| : S \in \mathcal{L}(V, W) \text{ and } \dim \text{range } S \leq k\} = s_{k+1}.$$

Furthermore, if

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m$$

is a singular value decomposition of T and $T_k \in \mathcal{L}(V, W)$ is defined by

$$T_k v = s_1 \langle v, e_1 \rangle f_1 + \dots + s_k \langle v, e_k \rangle f_k$$

for each $v \in V$, then $\dim \text{range } T_k = k$ and $\|T - T_k\| = s_{k+1}$.

Proof If $v \in V$ then

$$\begin{aligned} \|(T - T_k)v\|^2 &= \|s_{k+1} \langle v, e_{k+1} \rangle f_{k+1} + \dots + s_m \langle v, e_m \rangle f_m\|^2 \\ &= s_{k+1}^2 |\langle v, e_{k+1} \rangle|^2 + \dots + s_m^2 |\langle v, e_m \rangle|^2 \\ &\leq s_{k+1}^2 (|\langle v, e_{k+1} \rangle|^2 + \dots + |\langle v, e_m \rangle|^2) \\ &\leq s_{k+1}^2 \|v\|^2. \end{aligned}$$

Thus $\|T - T_k\| \leq s_{k+1}$. The equation $(T - T_k)e_{k+1} = s_{k+1}f_{k+1}$ now shows that $\|T - T_k\| = s_{k+1}$.

Suppose $S \in \mathcal{L}(V, W)$ and $\dim \text{range } S \leq k$. Thus $Se_1, \dots, Se_{k+1}$, which is a list of length $k + 1$, is linearly dependent. Hence there exist $a_1, \dots, a_{k+1} \in \mathbb{F}$, not all 0, such that

$$a_1 Se_1 + \dots + a_{k+1} Se_{k+1} = 0.$$

Now $a_1 e_1 + \dots + a_{k+1} e_{k+1} \neq 0$ because $a_1, \dots, a_{k+1}$ are not all 0. We have

$$\begin{aligned} \|(T - S)(a_1 e_1 + \dots + a_{k+1} e_{k+1})\|^2 &= \|T(a_1 e_1 + \dots + a_{k+1} e_{k+1})\|^2 \\ &= \|s_1 a_1 f_1 + \dots + s_{k+1} a_{k+1} f_{k+1}\|^2 \\ &= s_1^2 |a_1|^2 + \dots + s_{k+1}^2 |a_{k+1}|^2 \\ &\geq s_{k+1}^2 (|a_1|^2 + \dots + |a_{k+1}|^2) \\ &= s_{k+1}^2 \|a_1 e_1 + \dots + a_{k+1} e_{k+1}\|^2. \end{aligned}$$

Because $a_1 e_1 + \dots + a_{k+1} e_{k+1} \neq 0$, the inequality above implies that

$$\|T - S\| \geq s_{k+1}.$$

Thus $S = T_k$ minimizes $\|T - S\|$ among $S \in \mathcal{L}(V, W)$ with $\dim \text{range } S \leq k$. ■

For other examples of the use of the singular value decomposition in best approximation, see Exercise 22, which finds a subspace of given dimension on which the restriction of a linear map is as small as possible, and Exercise 27, which finds a unitary operator that is as close as possible to a given operator.

Polar Decomposition

Recall our discussion before 7.54 of the analogy between complex numbers z with $|z| = 1$ and unitary operators. Continuing with this analogy, note that every complex number z except 0 can be written in the form

$$\begin{aligned} z &= \left(\frac{z}{|z|} \right) |z| \\ &= \left(\frac{z}{|z|} \right) \sqrt{\bar{z}z}, \end{aligned}$$

where the first factor, namely, $z/|z|$, has absolute value 1.

Our analogy leads us to guess that every operator $T \in \mathcal{L}(V)$ can be written as a unitary operator times $\sqrt{T^*T}$. That guess is indeed correct. The corresponding result is called the polar decomposition, which gives a beautiful description of an arbitrary operator on V .

Note that if $T \in \mathcal{L}(V)$, then T^*T is a positive operator [as was shown in 7.64(a)]. Thus the operator $\sqrt{T^*T}$ makes sense and is well defined as a positive operator on V .

The polar decomposition that we are about to state and prove says that every operator on V is the product of a unitary operator and a positive operator. Thus we can write an arbitrary operator on V as the product of two nice operators, each of which comes from a class that we can completely describe and that we understand reasonably well. The unitary operators are described by 7.55 if $\mathbf{F} = \mathbf{C}$; the positive operators are described by the real and complex spectral theorems (7.29 and 7.31).

Specifically, consider the case $\mathbf{F} = \mathbf{C}$, and suppose

$$T = S\sqrt{T^*T}$$

is a polar decomposition of an operator $T \in \mathcal{L}(V)$, where S is a unitary operator. Then there is an orthonormal basis of V with respect to which S has a diagonal matrix, and there is an orthonormal basis of V with respect to which $\sqrt{T^*T}$ has a diagonal matrix. **Warning:** There may not exist an orthonormal basis that simultaneously puts the matrices of both S and $\sqrt{T^*T}$ into these nice diagonal forms— S may require one orthonormal basis and $\sqrt{T^*T}$ may require a different orthonormal basis.

However (still assuming that $\mathbf{F} = \mathbf{C}$), if T is normal, then an orthonormal basis of V can be chosen such that both S and $\sqrt{T^*T}$ have diagonal matrices with respect to this basis—see Exercise 31. The converse is also true: If $T \in \mathcal{L}(V)$ and $T = S\sqrt{T^*T}$ for some unitary operator $S \in \mathcal{L}(V)$ such that S and $\sqrt{T^*T}$ both have diagonal matrices with respect to the same orthonormal basis of V , then T is normal. This holds because T then has a diagonal matrix with respect to this same orthonormal basis, which implies that T is normal [by the equivalence of (c) and (a) in 7.31].

The polar decomposition below is valid on both real and complex inner product spaces and for all operators on those spaces.

7.93 polar decomposition

Suppose $T \in \mathcal{L}(V)$. Then there exists a unitary operator $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T}.$$

Proof Let $s_1, \dots, s_m$ be the positive singular values of T , and let $e_1, \dots, e_m$ and $f_1, \dots, f_m$ be orthonormal lists in V such that

$$7.94 \quad Tv = s_1\langle v, e_1 \rangle f_1 + \dots + s_m\langle v, e_m \rangle f_m$$

for every $v \in V$. Extend $e_1, \dots, e_m$ and $f_1, \dots, f_m$ to orthonormal bases $e_1, \dots, e_n$ and $f_1, \dots, f_n$ of V .

Define $S \in \mathcal{L}(V)$ by

$$Sv = \langle v, e_1 \rangle f_1 + \dots + \langle v, e_n \rangle f_n$$

for each $v \in V$. Then

$$\begin{aligned} \|Sv\|^2 &= \|\langle v, e_1 \rangle f_1 + \dots + \langle v, e_n \rangle f_n\|^2 \\ &= |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2 \\ &= \|v\|^2. \end{aligned}$$

Thus S is a unitary operator.

Applying T^* to both sides of 7.94 and then using the formula for T^* given by 7.77 shows that

$$T^*Tv = s_1^2\langle v, e_1 \rangle e_1 + \dots + s_m^2\langle v, e_m \rangle e_m$$

for every $v \in V$. Thus if $v \in V$, then

$$\sqrt{T^*T}v = s_1\langle v, e_1 \rangle e_1 + \dots + s_m\langle v, e_m \rangle e_m$$

because the operator that sends v to the right side of the equation above is a positive operator whose square equals T^*T . Now

$$\begin{aligned} S\sqrt{T^*T}v &= S(s_1\langle v, e_1 \rangle e_1 + \dots + s_m\langle v, e_m \rangle e_m) \\ &= s_1\langle v, e_1 \rangle f_1 + \dots + s_m\langle v, e_m \rangle f_m \\ &= Tv, \end{aligned}$$

where the last equation follows from 7.94. ■

Exercise 27 shows that the unitary operator S produced in the proof above is as close as a unitary operator can be to T .

Alternative proofs of the polar decomposition directly use the spectral theorem, avoiding the singular value decomposition. However, the proof above seems cleaner than those alternative proofs.

Operators Applied to Ellipsoids and Parallelepipeds

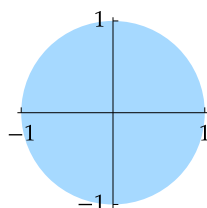
7.95 definition: ball, B

The *ball* in V of radius 1 centered at 0, denoted by B , is defined by

$$B = \{v \in V : \|v\| < 1\}.$$

If $\dim V = 2$, the word *disk* is sometimes used instead of *ball*. However, using *ball* in all dimensions is less confusing. Similarly, if $\dim V = 2$, then the word *ellipse* is sometimes used instead of the word *ellipsoid* that we are about to define. Again, using *ellipsoid* in all dimensions is less confusing.

You can think of the ellipsoid defined below as obtained by starting with the ball B and then stretching by a factor of s_k along each f_k -axis.



The ball B in $\mathbf{R}^2$.

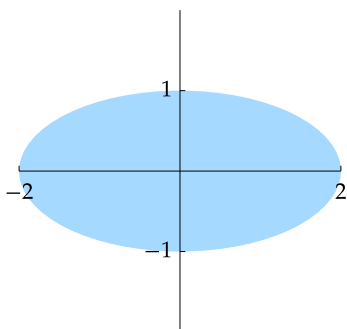
7.96 definition: ellipsoid, $E(s_1 f_1, \dots, s_n f_n)$, principal axes

Suppose that $f_1, \dots, f_n$ is an orthonormal basis of V and $s_1, \dots, s_n$ are positive numbers. The *ellipsoid* $E(s_1 f_1, \dots, s_n f_n)$ with *principal axes* $s_1 f_1, \dots, s_n f_n$ is defined by

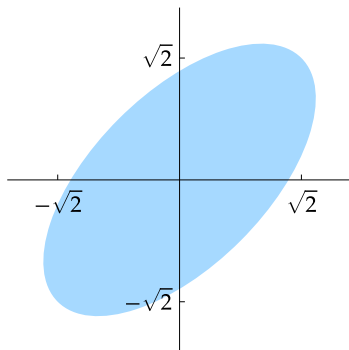
$$E(s_1 f_1, \dots, s_n f_n) = \left\{ v \in V : \frac{|\langle v, f_1 \rangle|^2}{s_1^2} + \dots + \frac{|\langle v, f_n \rangle|^2}{s_n^2} < 1 \right\}.$$

The ellipsoid notation $E(s_1 f_1, \dots, s_n f_n)$ does not explicitly include the inner product space V , even though the definition above depends on V . However, the inner product space V should be clear from the context and also from the requirement that $f_1, \dots, f_n$ be an orthonormal basis of V .

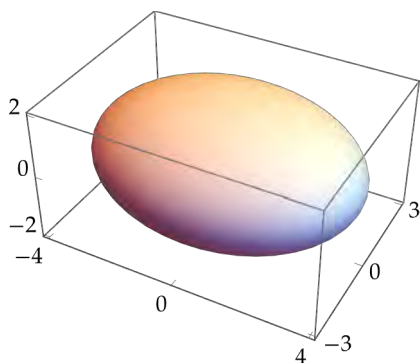
7.97 example: ellipsoids



The ellipsoid $E(2f_1, f_2)$ in $\mathbf{R}^2$, where f_1, f_2 is the standard basis of $\mathbf{R}^2$.



The ellipsoid $E(2f_1, f_2)$ in $\mathbf{R}^2$, where $f_1 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $f_2 = (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.



The ellipsoid $E(4f_1, 3f_2, 2f_3)$ in $\mathbb{R}^3$, where f_1, f_2, f_3 is the standard basis of $\mathbb{R}^3$.

The ellipsoid $E(f_1, \dots, f_n)$ equals the ball B in V for every orthonormal basis $f_1, \dots, f_n$ of V [by Parseval's identity 6.30(b)].

7.98 notation: $T(\Omega)$

For T a function defined on V and $\Omega \subseteq V$, define $T(\Omega)$ by

$$T(\Omega) = \{Tv : v \in \Omega\}.$$

Thus if T is a function defined on V , then $T(V) = \text{range } T$.

The next result states that every invertible operator $T \in \mathcal{L}(V)$ maps the ball B in V onto an ellipsoid in V . The proof shows that the principal axes of this ellipsoid come from the singular value decomposition of T .

7.99 invertible operator takes ball to ellipsoid

Suppose $T \in \mathcal{L}(V)$ is invertible. Then T maps the ball B in V onto an ellipsoid in V .

Proof Suppose T has singular value decomposition

$$7.100 \quad Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_n \langle v, e_n \rangle f_n$$

for all $v \in V$; here $s_1, \dots, s_n$ are the singular values of T and $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are both orthonormal bases of V . We will show that $T(B) = E(s_1 f_1, \dots, s_n f_n)$.

First suppose $v \in B$. Because T is invertible, none of the singular values $s_1, \dots, s_n$ equals 0 (see 7.68). Thus 7.100 implies that

$$\frac{|\langle Tv, f_1 \rangle|^2}{s_1^2} + \dots + \frac{|\langle Tv, f_n \rangle|^2}{s_n^2} = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2 < 1.$$

Thus $Tv \in E(s_1 f_1, \dots, s_n f_n)$. Hence $T(B) \subseteq E(s_1 f_1, \dots, s_n f_n)$.

To prove inclusion in the other direction, now suppose $w \in E(s_1 f_1, \dots, s_n f_n)$.

Let

$$v = \frac{\langle w, f_1 \rangle}{s_1} e_1 + \dots + \frac{\langle w, f_n \rangle}{s_n} e_n.$$

Then $\|v\| < 1$ and 7.100 implies that $Tv = \langle w, f_1 \rangle f_1 + \dots + \langle w, f_n \rangle f_n = w$. Thus $T(B) \supseteq E(s_1 f_1, \dots, s_n f_n)$. ■

We now use the previous result to show that invertible operators take all ellipsoids, not just the ball of radius 1, to ellipsoids.

7.101 *invertible operator takes ellipsoids to ellipsoids*

Suppose $T \in \mathcal{L}(V)$ is invertible and E is an ellipsoid in V . Then $T(E)$ is an ellipsoid in V .

Proof There exist an orthonormal basis $f_1, \dots, f_n$ of V and positive numbers $s_1, \dots, s_n$ such that $E = E(s_1 f_1, \dots, s_n f_n)$. Define $S \in \mathcal{L}(V)$ by

$$S(a_1 f_1 + \dots + a_n f_n) = a_1 s_1 f_1 + \dots + a_n s_n f_n.$$

Then S maps the ball B of V onto E , as you can verify. Thus

$$T(E) = T(S(B)) = (TS)(B).$$

The equation above and 7.99, applied to TS , show that $T(E)$ is an ellipsoid in V . ■

Recall (see 3.95) that if $u \in V$ and $\Omega \subseteq V$ then $u + \Omega$ is defined by

$$u + \Omega = \{u + w : w \in \Omega\}.$$

Geometrically, the sets Ω and $u + \Omega$ look the same, but they are in different locations.

In the following definition, if $\dim V = 2$ then the word *parallelogram* is often used instead of *parallelepiped*.

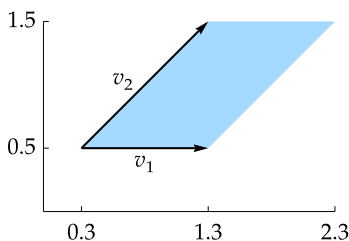
7.102 *definition: $P(v_1, \dots, v_n)$, parallelepiped*

Suppose $v_1, \dots, v_n$ is a basis of V . Let

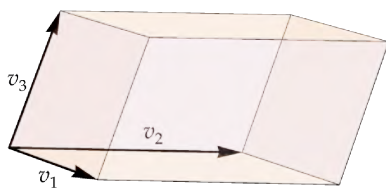
$$P(v_1, \dots, v_n) = \{a_1 v_1 + \dots + a_n v_n : a_1, \dots, a_n \in (0, 1)\}.$$

A *parallelepiped* is a set of the form $u + P(v_1, \dots, v_n)$ for some $u \in V$. The vectors $v_1, \dots, v_n$ are called the *edges* of this parallelepiped.

7.103 *example: parallelepipeds*



The parallelepiped
 $(0.3, 0.5) + P((1, 0), (1, 1))$ in $\mathbb{R}^2$.



A parallelepiped in $\mathbb{R}^3$.

7.104 invertible operator takes parallelepipeds to parallelepipeds

Suppose $u \in V$, $v_1, \dots, v_n$ is a basis of V , and $T \in \mathcal{L}(V)$ is invertible. Then

$$T(u + P(v_1, \dots, v_n)) = Tu + P(Tv_1, \dots, Tv_n).$$

Proof Because T is invertible, the list $Tv_1, \dots, Tv_n$ is a basis of V . The linearity of T implies that

$$T(u + a_1v_1 + \dots + a_nv_n) = Tu + a_1Tv_1 + \dots + a_nTv_n$$

for all $a_1, \dots, a_n \in (0, 1)$. Thus $T(u + P(v_1, \dots, v_n)) = Tu + P(Tv_1, \dots, Tv_n)$. ■

Just as the rectangles are distinguished among the parallelograms in $\mathbf{R}^2$, we give a special name to the parallelepipeds in V whose defining edges are orthogonal to each other.

7.105 definition: box

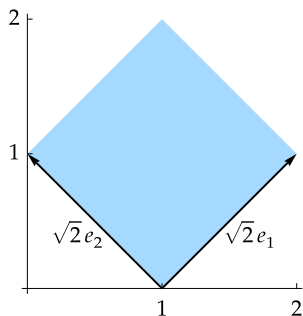
A box in V is a set of the form

$$u + P(r_1e_1, \dots, r_ne_n),$$

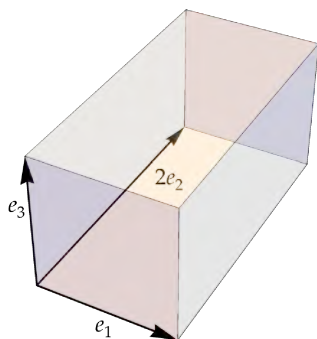
where $u \in V$ and $r_1, \dots, r_n$ are positive numbers and $e_1, \dots, e_n$ is an orthonormal basis of V .

Note that in the special case of $\mathbf{R}^2$ each box is a rectangle, but the terminology *box* can be used in all dimensions.

7.106 example: boxes



The box $(1, 0) + P(\sqrt{2}e_1, \sqrt{2}e_2)$, where $e_1 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $e_2 = (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.



The box $P(e_1, 2e_2, e_3)$, where e_1, e_2, e_3 is the standard basis of $\mathbf{R}^3$.

Suppose $T \in \mathcal{L}(V)$ is invertible. Then T maps every parallelepiped in V to a parallelepiped in V (by 7.104). In particular, T maps every box in V to a parallelepiped in V . This raises the question of whether T maps some boxes in V to boxes in V . The following result answers this question, with the help of the singular value decomposition.

7.107 every invertible operator takes some boxes to boxes

Suppose $T \in \mathcal{L}(V)$ is invertible. Suppose T has singular value decomposition

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_n \langle v, e_n \rangle f_n,$$

where $s_1, \dots, s_n$ are the singular values of T and $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are orthonormal bases of V and the equation above holds for all $v \in V$. Then T maps the box $u + P(r_1 e_1, \dots, r_n e_n)$ onto the box $Tu + P(r_1 s_1 f_1, \dots, r_n s_n f_n)$ for all positive numbers $r_1, \dots, r_n$ and all $u \in V$.

Proof If $a_1, \dots, a_n \in (0, 1)$ and $r_1, \dots, r_n$ are positive numbers and $u \in V$, then

$$T(u + a_1 r_1 e_1 + \cdots + a_n r_n e_n) = Tu + a_1 r_1 s_1 f_1 + \cdots + a_n r_n s_n f_n.$$

Thus $T(u + P(r_1 e_1, \dots, r_n e_n)) = Tu + P(r_1 s_1 f_1, \dots, r_n s_n f_n)$. ■

Volume via Singular Values

Our goal in this subsection is to understand how an operator changes the volume of subsets of its domain. Because notions of volume belong to analysis rather than to linear algebra, we will work only with an intuitive notion of volume. Our intuitive approach to volume can be converted into appropriate correct definitions, correct statements, and correct proofs using the machinery of analysis.

Our intuition about volume works best in real inner product spaces. Thus the assumption that $\mathbf{F} = \mathbf{R}$ will appear frequently in the rest of this subsection.

If $\dim V = n$, then by *volume* we will mean n -dimensional volume. You should be familiar with this concept in $\mathbf{R}^3$. When $n = 2$, this is usually called area instead of volume, but for consistency we use the word volume in all dimensions. The most fundamental intuition about volume is that the volume of a box (whose defining edges are by definition orthogonal to each other) is the product of the lengths of the defining edges. Thus we make the following definition.

7.108 definition: volume of a box

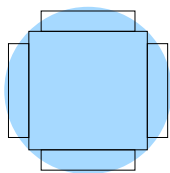
Suppose $\mathbf{F} = \mathbf{R}$. If $u \in V$ and $r_1, \dots, r_n$ are positive numbers and $e_1, \dots, e_n$ is an orthonormal basis of V , then

$$\text{volume}(u + P(r_1 e_1, \dots, r_n e_n)) = r_1 \times \cdots \times r_n.$$

The definition above agrees with the familiar formulas for the area (which we are calling the volume) of a rectangle in $\mathbf{R}^2$ and for the volume of a box in $\mathbf{R}^3$. For example, the first box in Example 7.106 has two-dimensional volume (or area) 2 because the defining edges of that box have length $\sqrt{2}$ and $\sqrt{2}$. The second box in Example 7.106 has three-dimensional volume 2 because the defining edges of that box have length 1, 2, and 1.

To define the volume of a subset of V , approximate the subset by a finite collection of disjoint boxes, and then add up the volumes of the approximating collection of boxes. As we approximate a subset of V more accurately by disjoint unions of more boxes, we get a better approximation to the volume.

These ideas should remind you of how the Riemann integral is defined by approximating the area under a curve by a disjoint collection of rectangles. This discussion leads to the following nonrigorous but intuitive definition.



Volume of this ball $\approx$ sum of the volumes of the five boxes.

7.109 definition: *volume*

Suppose $\mathbf{F} = \mathbf{R}$ and $\Omega \subseteq V$. Then the *volume* of Ω , denoted by $\text{volume } \Omega$, is approximately the sum of the volumes of a collection of disjoint boxes that approximate Ω .

We are ignoring many reasonable questions by taking an intuitive approach to volume. For example, if we approximate Ω by boxes with respect to one basis, do we get the same volume if we approximate Ω by boxes with respect to a different basis? If Ω_1 and Ω_2 are disjoint subsets of V , is $\text{volume}(\Omega_1 \cup \Omega_2) = \text{volume } \Omega_1 + \text{volume } \Omega_2$? Provided that we consider only reasonably nice subsets of V , techniques of analysis show that both these questions have affirmative answers that agree with our intuition about volume.

7.110 example: *volume change by a linear map*

Suppose that $T \in \mathcal{L}(\mathbf{R}^2)$ is defined by $Tv = 2\langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2$, where e_1, e_2 is the standard basis of $\mathbf{R}^2$. This linear map stretches vectors along the e_1 -axis by a factor of 2 and leaves vectors along the e_2 -axis unchanged. The ball approximated by five boxes above gets mapped by T to the ellipsoid shown here. Each of the five boxes in the original figure gets mapped to a box of twice the width and the same height as in the original figure. Hence each box gets mapped to a box of twice the volume (area) as in the original figure. The sum of the volumes of the five new boxes approximates the volume of the ellipsoid. Thus T changes the volume of the ball by a factor of 2.



Each box here has twice the width and the same height as the boxes in the previous figure.

In the example above, T maps boxes with respect to the basis e_1, e_2 to boxes with respect to the same basis; thus we can see how T changes volume. In general, an operator maps boxes to parallelepipeds that are not boxes. However, if we choose the right basis (coming from the singular value decomposition!), then boxes with respect to that basis get mapped to boxes with respect to a possibly different basis, as shown in 7.107. This observation leads to a natural proof of the following result.

7.111 *volume changes by a factor of the product of the singular values*

Suppose $\mathbf{F} = \mathbf{R}$, $T \in \mathcal{L}(V)$ is invertible, and $\Omega \subseteq V$. Then

$$\text{volume } T(\Omega) = (\text{product of singular values of } T)(\text{volume } \Omega).$$

Proof Suppose T has singular value decomposition

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_n \langle v, e_n \rangle f_n$$

for all $v \in V$, where $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are orthonormal bases of V .

Approximate Ω by boxes of the form $u + P(r_1 e_1, \dots, r_n e_n)$, which have volume $r_1 \times \cdots \times r_n$. The operator T maps each box $u + P(r_1 e_1, \dots, r_n e_n)$ onto the box $Tu + P(r_1 s_1 f_1, \dots, r_n s_n f_n)$, which has volume $(s_1 \times \cdots \times s_n)(r_1 \times \cdots \times r_n)$.

The operator T maps a collection of boxes that approximate Ω onto a collection of boxes that approximate $T(\Omega)$. Because T changes the volume of each box in a collection that approximates Ω by a factor of $s_1 \times \cdots \times s_n$, the linear map T changes the volume of Ω by the same factor. ■

Suppose $T \in \mathcal{L}(V)$. As we will see when we get to determinants, the product of the singular values of T equals $|\det T|$; see 9.60 and 9.61.

Properties of an Operator as Determined by Its Eigenvalues

We conclude this chapter by presenting the table below. The context of this table is a finite-dimensional complex inner product space. The first column of the table shows a property that a normal operator on such a space might have. The second column of the table shows a subset of $\mathbf{C}$ such that the operator has the corresponding property if and only if all eigenvalues of the operator lie in the specified subset. For example, the first row of the table states that a normal operator is invertible if and only if all its eigenvalues are nonzero (this first row is the only one in the table that does not need the hypothesis that the operator is normal).

Make sure you can explain why all results in the table hold. For example, the last row of the table holds because the norm of an operator equals its largest singular value (by 7.85) and the singular values of a normal operator, assuming $\mathbf{F} = \mathbf{C}$, equal the absolute values of the eigenvalues (by Exercise 7 in Section 7E).

properties of a normal operator	eigenvalues are contained in
invertible	$\mathbf{C} \setminus \{0\}$
self-adjoint	$\mathbf{R}$
skew	$\{\lambda \in \mathbf{C} : \operatorname{Re} \lambda = 0\}$
orthogonal projection	$\{0, 1\}$
positive	$[0, \infty)$
unitary	$\{\lambda \in \mathbf{C} : \lambda = 1\}$
norm is less than 1	$\{\lambda \in \mathbf{C} : \lambda < 1\}$

Exercises 7F

- 1 Prove that if $S, T \in \mathcal{L}(V, W)$, then $|\|S\| - \|T\|| \leq \|S - T\|$.

*The inequality above is called the **reverse triangle inequality**.*

- 2 Suppose that $T \in \mathcal{L}(V)$ is self-adjoint or that $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that

$$\|T\| = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } T\}.$$

- 3 Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Prove that

$$\|Tv\| = \|T\| \|v\| \iff T^*Tv = \|T\|^2 v.$$

- 4 Suppose $T \in \mathcal{L}(V, W)$, $v \in V$, and $\|Tv\| = \|T\| \|v\|$. Prove that if $u \in V$ and $\langle u, v \rangle = 0$, then $\langle Tu, Tv \rangle = 0$.

- 5 Suppose U is a finite-dimensional inner product space, $T \in \mathcal{L}(V, U)$, and $S \in \mathcal{L}(U, W)$. Prove that

$$\|ST\| \leq \|S\| \|T\|.$$

- 6 Prove or give a counterexample: If $S, T \in \mathcal{L}(V)$, then $\|ST\| = \|TS\|$.

- 7 Show that defining $d(S, T) = \|S - T\|$ for $S, T \in \mathcal{L}(V, W)$ makes d a metric on $\mathcal{L}(V, W)$.

This exercise is intended for readers who are familiar with metric spaces.

- 8 (a) Prove that if $T \in \mathcal{L}(V)$ and $\|I - T\| < 1$, then T is invertible.
 (b) Suppose that $S \in \mathcal{L}(V)$ is invertible. Prove that if $T \in \mathcal{L}(V)$ and $\|S - T\| < 1/\|S^{-1}\|$, then T is invertible.

This exercise shows that the set of invertible operators in $\mathcal{L}(V)$ is an open subset of $\mathcal{L}(V)$, using the metric defined in Exercise 7.

- 9 Suppose $T \in \mathcal{L}(V)$. Prove that for every $\epsilon > 0$, there exists an invertible operator $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \epsilon$.
- 10 Suppose $\dim V > 1$ and $T \in \mathcal{L}(V)$ is not invertible. Prove that for every $\epsilon > 0$, there exists $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \epsilon$ and S is not invertible.
- 11 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that for every $\epsilon > 0$ there exists a diagonalizable operator $S \in \mathcal{L}(V)$ such that $0 < \|T - S\| < \epsilon$.

- 12 Suppose $T \in \mathcal{L}(V)$ is a positive operator. Show that $\|\sqrt{T}\| = \sqrt{\|T\|}$.

- 13 Suppose $S, T \in \mathcal{L}(V)$ are positive operators. Show that

$$\|S - T\| \leq \max\{\|S\|, \|T\|\} \leq \|S + T\|.$$

- 14 Suppose U and W are subspaces of V such that $\|P_U - P_W\| < 1$. Prove that $\dim U = \dim W$.

- 15 Define $T \in \mathcal{L}(\mathbf{F}^3)$ by

$$T(z_1, z_2, z_3) = (z_3, 2z_1, 3z_2).$$

Find (explicitly) a unitary operator $S \in \mathcal{L}(\mathbf{F}^3)$ such that $T = S\sqrt{T^*T}$.

- 16 Suppose $S \in \mathcal{L}(V)$ is a positive invertible operator. Prove that there exists $\delta > 0$ such that T is a positive operator for every self-adjoint operator $T \in \mathcal{L}(V)$ with $\|S - T\| < \delta$.

- 17 Prove that if $u \in V$ and φ_u is the linear functional on V defined by the equation $\varphi_u(v) = \langle v, u \rangle$, then $\|\varphi_u\| = \|u\|$.

Here we are thinking of the scalar field $\mathbf{F}$ as an inner product space with $\langle \alpha, \beta \rangle = \alpha \bar{\beta}$ for all $\alpha, \beta \in \mathbf{F}$. Thus $\|\varphi_u\|$ means the norm of φ_u as a linear map from V to $\mathbf{F}$.

- 18 Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $T \in \mathcal{L}(V, W)$.

(a) Prove that $\max\{\|Te_1\|, \dots, \|Te_n\|\} \leq \|T\| \leq (\|Te_1\|^2 + \dots + \|Te_n\|^2)^{1/2}$.

(b) Prove that $\|T\| = (\|Te_1\|^2 + \dots + \|Te_n\|^2)^{1/2}$ if and only if $\dim \text{range } T \leq 1$.

Here $e_1, \dots, e_n$ is an arbitrary orthonormal basis of V , not necessarily connected with a singular value decomposition of T . If $s_1, \dots, s_n$ is the list of singular values of T , then the right side of the inequality above equals $(s_1^2 + \dots + s_n^2)^{1/2}$, as was shown in Exercise 11(a) in Section 7E.

- 19 Prove that if $T \in \mathcal{L}(V, W)$, then $\|T^*T\| = \|T\|^2$.

*This formula for $\|T^*T\|$ leads to the important subject of C^* -algebras.*

- 20 Suppose $T \in \mathcal{L}(V)$ is normal. Prove that $\|T^k\| = \|T\|^k$ for every positive integer k .

- 21 Suppose $\dim V > 1$ and $\dim W > 1$. Prove that the norm on $\mathcal{L}(V, W)$ does not come from an inner product. In other words, prove that there does not exist an inner product on $\mathcal{L}(V, W)$ such that

$$\max\{\|Tv\| : v \in V \text{ and } \|v\| \leq 1\} = \sqrt{\langle T, T \rangle}$$

for all $T \in \mathcal{L}(V, W)$.

- 22 Suppose $T \in \mathcal{L}(V, W)$. Let $n = \dim V$ and let $s_1 \geq \dots \geq s_n$ denote the singular values of T . Prove that if $1 \leq k \leq n$, then

$$\min\{\|T|_U\| : U \text{ is a subspace of } V \text{ with } \dim U = k\} = s_{n-k+1}.$$

- 23 Suppose $T \in \mathcal{L}(V, W)$. Show that T is uniformly continuous with respect to the metrics on V and W that arise from the norms on those spaces (see Exercise 23 in Section 6B).

- 24 Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$\|T^{-1}\| = \|T\|^{-1} \iff \frac{T}{\|T\|} \text{ is a unitary operator.}$$

- 25** Fix $u, x \in V$ with $u \neq 0$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u \rangle x$ for every $v \in V$. Prove that

$$\sqrt{T^*T}v = \frac{\|x\|}{\|u\|} \langle v, u \rangle u$$

for every $v \in V$.

- 26** Suppose $T \in \mathcal{L}(V)$. Prove that T is invertible if and only if there exists a unique unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$.
- 27** Suppose $T \in \mathcal{L}(V)$ and $s_1, \dots, s_n$ are the singular values of T . Let $e_1, \dots, e_n$ and $f_1, \dots, f_n$ be orthonormal bases of V such that

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_n \langle v, e_n \rangle f_n$$

for all $v \in V$. Define $S \in \mathcal{L}(V)$ by

$$Sv = \langle v, e_1 \rangle f_1 + \dots + \langle v, e_n \rangle f_n.$$

- (a) Show that S is unitary and $\|T - S\| = \max\{|s_1 - 1|, \dots, |s_n - 1|\}$.
 (b) Show that if $E \in \mathcal{L}(V)$ is unitary, then $\|T - E\| \geq \|T - S\|$.

This exercise finds a unitary operator S that is as close as possible (among the unitary operators) to a given operator T .

- 28** Suppose $T \in \mathcal{L}(V)$. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = \sqrt{TT^*}S$.
- 29** Suppose $T \in \mathcal{L}(V)$.
- (a) Use the polar decomposition to show that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $TT^* = ST^*TS^*$.
 (b) Show how (a) implies that T and T^* have the same singular values.

- 30** Suppose $T \in \mathcal{L}(V)$, $S \in \mathcal{L}(V)$ is a unitary operator, and $R \in \mathcal{L}(V)$ is a positive operator such that $T = SR$. Prove that $R = \sqrt{T^*T}$.

*This exercise shows that if we write T as the product of a unitary operator and a positive operator (as in the polar decomposition 7.93), then the positive operator equals $\sqrt{T^*T}$.*

- 31** Suppose $F = \mathbb{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that there exists a unitary operator $S \in \mathcal{L}(V)$ such that $T = S\sqrt{T^*T}$ and such that S and $\sqrt{T^*T}$ both have diagonal matrices with respect to the same orthonormal basis of V .
- 32** Suppose that $T \in \mathcal{L}(V, W)$ and $T \neq 0$. Let $s_1, \dots, s_m$ denote the positive singular values of T . Show that there exists an orthonormal basis $e_1, \dots, e_m$ of $(\text{null } T)^\perp$ such that

$$T\left(E\left(\frac{e_1}{s_1}, \dots, \frac{e_m}{s_m}\right)\right)$$

equals the ball in $\text{range } T$ of radius 1 centered at 0.